

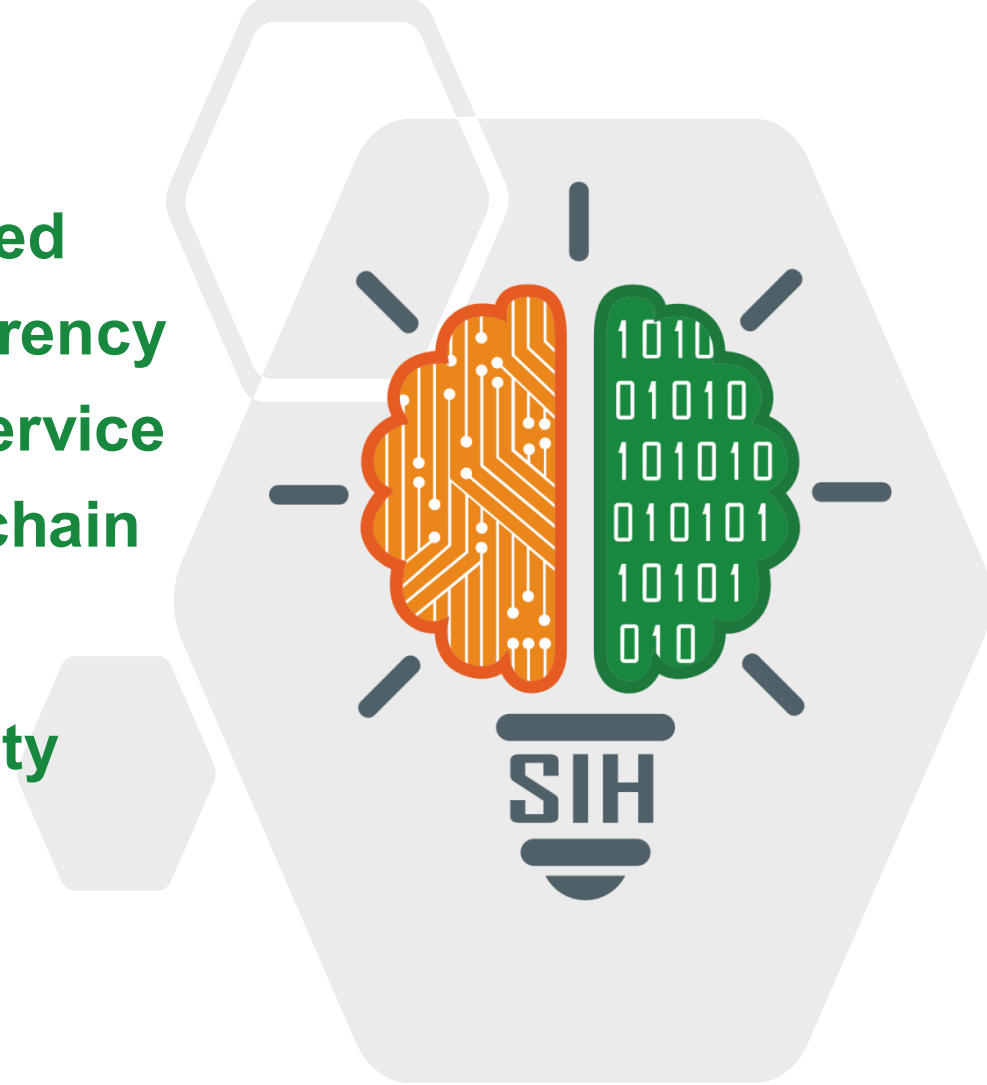


SMART INDIA HACKATHON 2026



SMART INDIA
HACKATHON
2026

- Problem Statement ID – **SIH26182**
- Problem Statement Title- **Automated Attribution of Unknown Cryptocurrency Wallets to Nearest Virtual Asset Service Providers (VASPs) through Blockchain Intelligence APIs**
- Theme- **Blockchain & Cybersecurity**
- PS Category- **Software**
- Team ID-
- Team Name (Registered on portal)- **Achievers**



Problem Existing

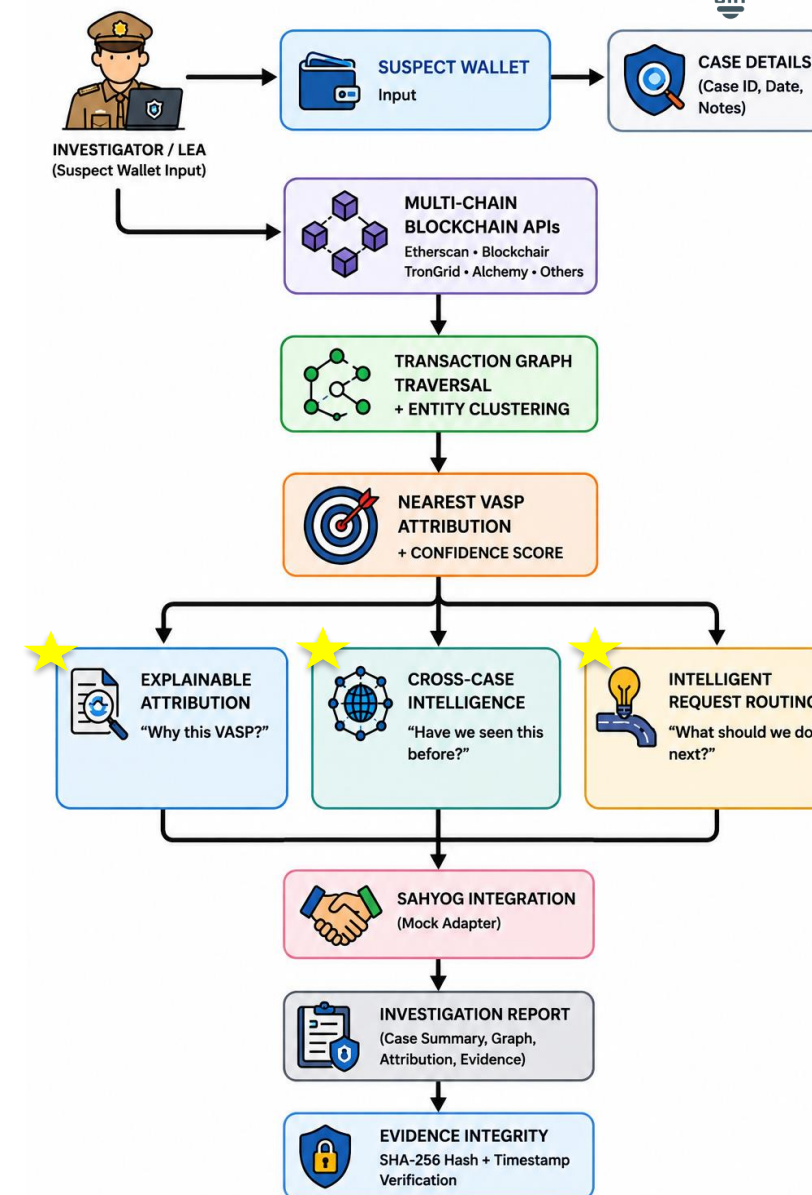
Imagine a cybercrime victim loses ₹50,000, which is converted into cryptocurrency and moved across different blockchain networks and services, before reaching a crypto exchange. The Law Enforcement Agencies (LEAs) can see the wallet addresses and transactions, but cannot directly tell who owns the wallets or which Virtual Asset Service Provider (VASP)/exchange ultimately received the funds. Tracing this flow manually across multiple blockchains, wallets and bridges is slow and complex. This creates major delays in attribution, freezing of assets, and identification of the beneficial owner.

Proposed Solution

- **Automated Multi-chain Tracing:** Analyse suspect wallets and trace transactions across major blockchains.
- **Entity Identification:** Identify and tagging of wallets, clusters, deposits, mixers, bridges and cross chain swap services.
- **VASP Attribution:** Find nearest VASP linked to unknown wallets with confidence scoring.
- **Graph Analysis:** Visualise transaction paths and fund flows.
- **SAHYOG Integration:** Generate reports and assist investigators in routing lawful disclosure.

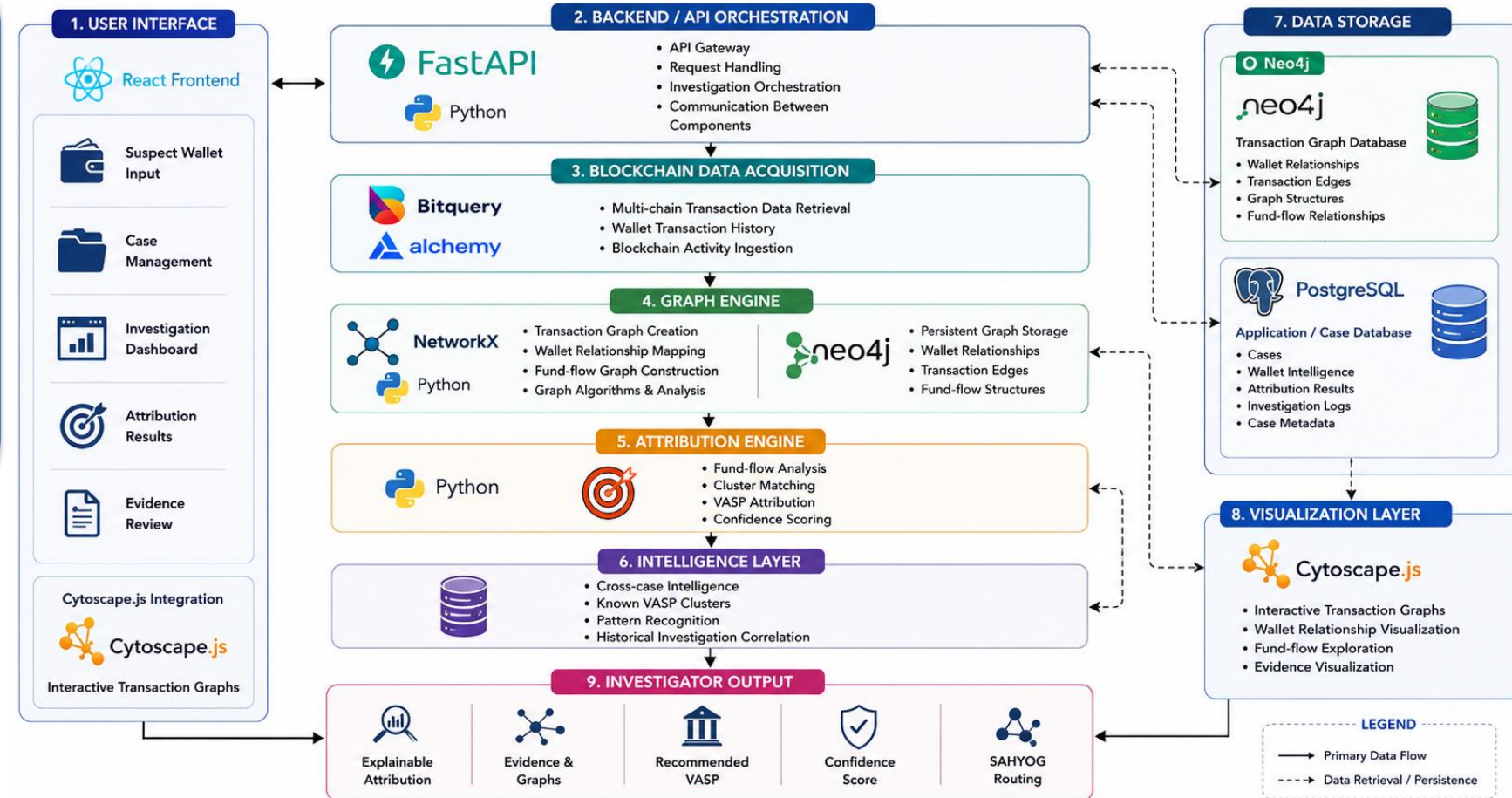
UVP (Unique Value Proposition)

Innovation	Key Value
➤ Explainable Attribution	Provides evidence for each VASP attribution through transaction hops, cluster matches, fund flow analysis, and confidence scoring.
➤ Cross-Case Intelligence	Connects wallets and transaction infrastructure across previous cases to identify potentially recurring or shared patterns.
➤ Intelligent Request Routing	Recommends the appropriate next action—disclosure/KYC, freeze/preservation, or further tracing—based on investigation intelligence.
➤ Evidence Integrity	The investigator can check if the report was modified.



TECH STACK

- **MULTI-CHAIN APIs:** Bitquery and Alchemy provide blockchain transaction, wallet and fund-flow data across supported networks.
- **BACKEND ENGINE:** Python and FastAPI handle transaction processing, tracing logic, VASP attribution and system APIs.
- **GRAPH ANALYSIS:** NetworkX builds transaction networks and analyzes multi-hop fund flows and wallet relationships.
- **DATABASE:** PostgreSQL securely stores investigation cases, wallet intelligence, attribution results and historical records.
- **VISUALIZATION:** React.js and Cytoscape.js provide an interactive dashboard with visual transaction and wallet networks.



Technical Feasibility

- **Multi-Chain Blockchain APIs:** Access transaction data across major supported blockchains without running individual blockchain nodes.
- **Graph-Based Tracing:** Automatically analyze multi-hop fund movement and wallet relationships.
- **VASP Attribution Engine:** Match transaction paths and wallet clusters against known VASP intelligence to rank likely VASPs.
- **Modular Architecture:** Chain-specific adapters allow additional blockchain networks to be integrated easily.

Economic Feasibility

- **API-Based Infrastructure:** Uses existing blockchain APIs, avoiding the need to maintain blockchain nodes initially.
- **Open-Source Technology:** Uses open-source tools to reduce software licensing and development costs.
- **Pay-as-Scale Model:** API, cloud and storage resources can increase as investigation volume grows.
- **Reduced Investigation Effort:** Automates repetitive tracing, saving investigator time and speeding up VASP identification.

Attribution Feasibility

- **Known VASP Intelligence:** VASP associated addresses and wallet clusters as reliable attribution references.
- **Transaction-Path Analysis:** Multi-hop fund-flow analysis to identify connections with VASP-associated addresses.
- **Multi-Signal Analysis:** Fund flow, wallet relationships and transaction patterns for stronger attribution.
- **Confidence-Based Ranking:** Candidate VASPs ranked according to the strength of available blockchain evidence.
- **Explainable Attribution:** Transaction paths and supporting evidence behind every VASP attribution.



KEY CHALLENGES



MITIGATION



Large Transaction Graphs



Limit tracing by hops, time period and transaction value to focus on relevant flows.



Unknown Wallet Ownership



Combine wallet clusters, fund flows and transaction patterns instead of relying on a single indicator.



False VASP Attribution



Use confidence scores and show alternative VASP candidates when evidence is insufficient.



Cross-Chain Complexity



Use chain-specific adapters with a common data format for consistent analysis.



API Rate Limits



Use caching and multiple API providers to maintain reliable data

Direct Impact on LEAs

- **Faster Investigations:** Automates multi-chain transaction tracing and VASP identification.
- **Reduced Manual Effort:** Minimizes time spent analysing wallets, transactions and intermediary addresses.
- **Actionable Intelligence:** Provides VASP attribution, confidence scores and risk insights in one dashboard.
- **Better Case Investigation:** Connects current wallets with potentially recurring infrastructure from previous cases.

Operational & Strategic Impact

- **Faster Asset Freezing:** Enables quicker identification and routing of requests to relevant VASPs.
- **Multi-Chain Capability:** Supports investigation across major blockchain networks and services.
- **Explainable Decisions:** Provides evidence behind each VASP attribution for investigator review.
- **Scalable Intelligence:** Builds a growing cross-case network of wallet and transaction intelligence.

Overall Benefits

- | | |
|------------------------|-------------------|
| ➤ ↓ Investigation Time | ➤ ↑ Visibility |
| ➤ ↑ VASP Attribution | ➤ ↑ Faster Action |

Direct Impact on LEAs

- **Faster Investigations:** Automates multi-chain transaction tracing and VASP identification.
- **Reduced Manual Effort:** Minimizes time spent analysing wallets, transactions and intermediary addresses.
- **Actionable Intelligence:** Provides VASP attribution, confidence scores and risk insights in one dashboard.
- **Better Case Investigation:** Connects current wallets with potentially recurring infrastructure from previous cases.

Operational & Strategic Impact

- **Faster Asset Freezing:** Enables quicker identification and routing of requests to relevant VASPs.
- **Multi-Chain Capability:** Supports investigation across major blockchain networks and services.
- **Explainable Decisions:** Provides evidence behind each VASP attribution for investigator review.
- **Scalable Intelligence:** Builds a growing cross-case network of wallet and transaction intelligence.

Overall Benefits

- | | |
|------------------------|-------------------|
| ➤ ↓ Investigation Time | ➤ ↑ Visibility |
| ➤ ↑ VASP Attribution | ➤ ↑ Faster Action |

RESEARCH AND REFERENCES

- Details / Links of the reference and research work

Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.